LLMs Don't Talk Like Humans (Yet): Measuring the Authenticity Gap in Social Media Simulations

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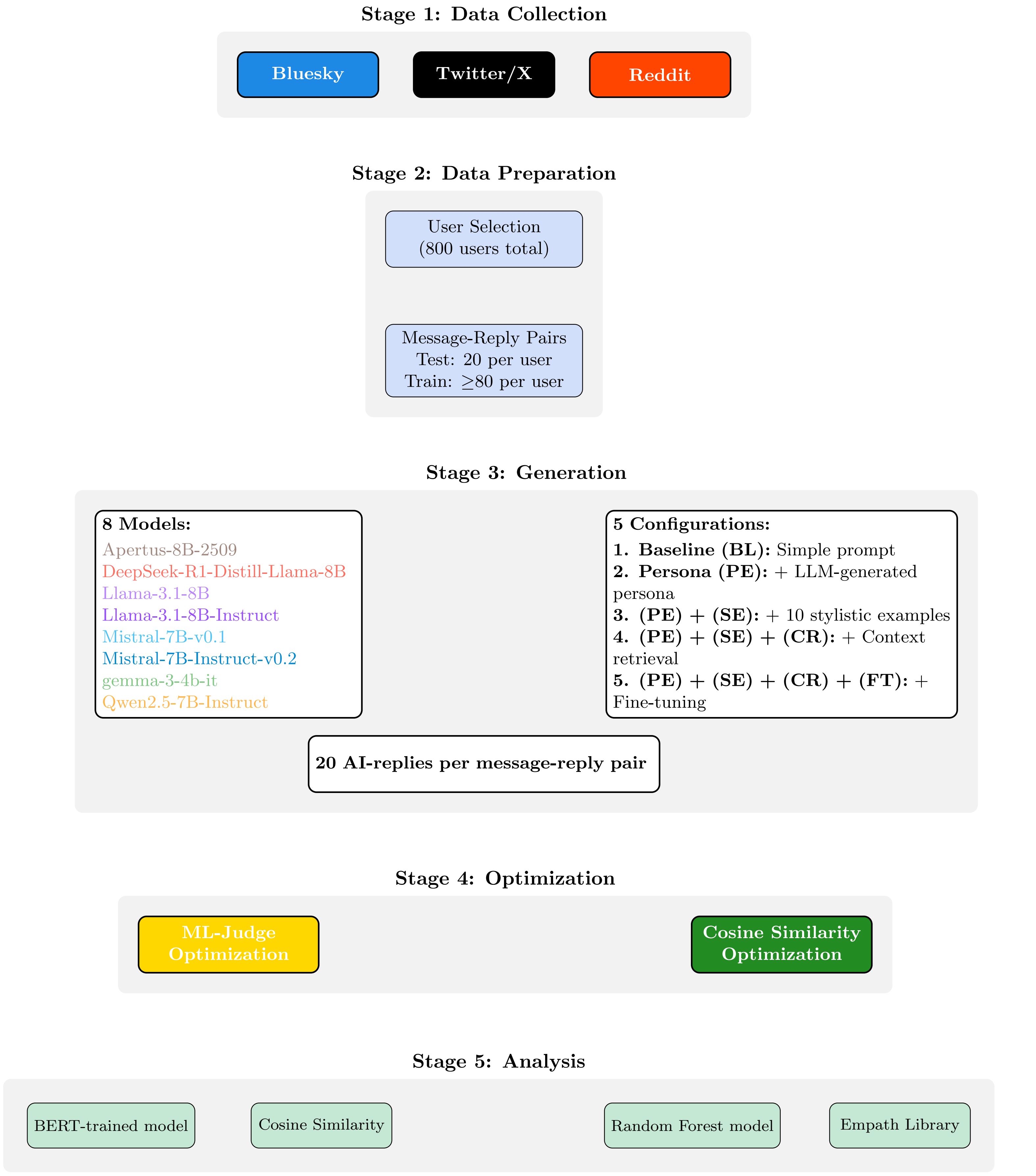
MOTIVATION:

Social scientists increasingly deploy LLMs as stand-ins for human participants—generating synthetic survey responses, simulating experimental subjects, and creating social media personas. But there is a **critical assumption** underlying all these applications: that **LLM output is indistinguishable from genuine human communication. Is it?**

VALIDATION FRAMEWORK & CALIBRATION:



Stage 1: Data Collection

Bluesky | Twitter/X | Reddit

Stage 2: Data Preparation

User Selection
(800 users total)

Message-Reply Pairs
Test: 20 per user
Train: ≥80 per user

Stage 3: Generation

8 Models:
- Apertus-8B-2509
- DeepSeek-R1-Distill-Llama-8B
- Llama-3.1-8B
- Llama-3.1-8B-Instruct
- Mistral-7B-v0.1
- Mistral-7B-Instruct-v0.2
- gemma-3-4b-it
- Qwen2.5-7B-Instruct

5 Configurations:
1. **Baseline (BL):** Simple prompt
2. **Persona (PE):** + LLM-generated persona
3. **(PE) + (SE):** + 10 stylistic examples
4. **(PE) + (SE) + (CR):** + Context retrieval
5. **(PE) + (SE) + (CR) + (FT):** + Fine-tuning

20 AI-replies per message-reply pair

Stage 4: Optimization

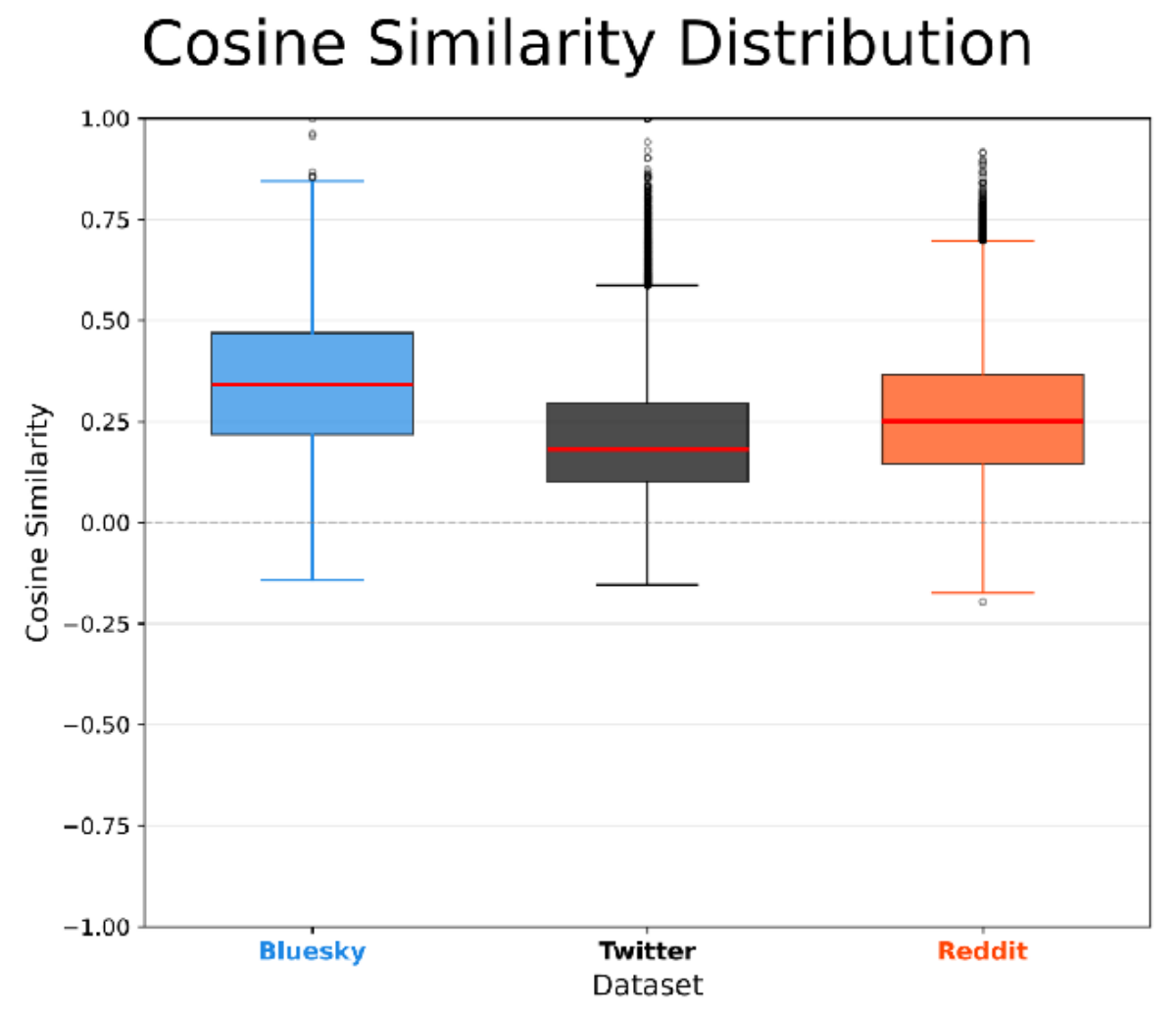
ML-Judge Optimization | Cosine Similarity Optimization

Stage 5: Analysis

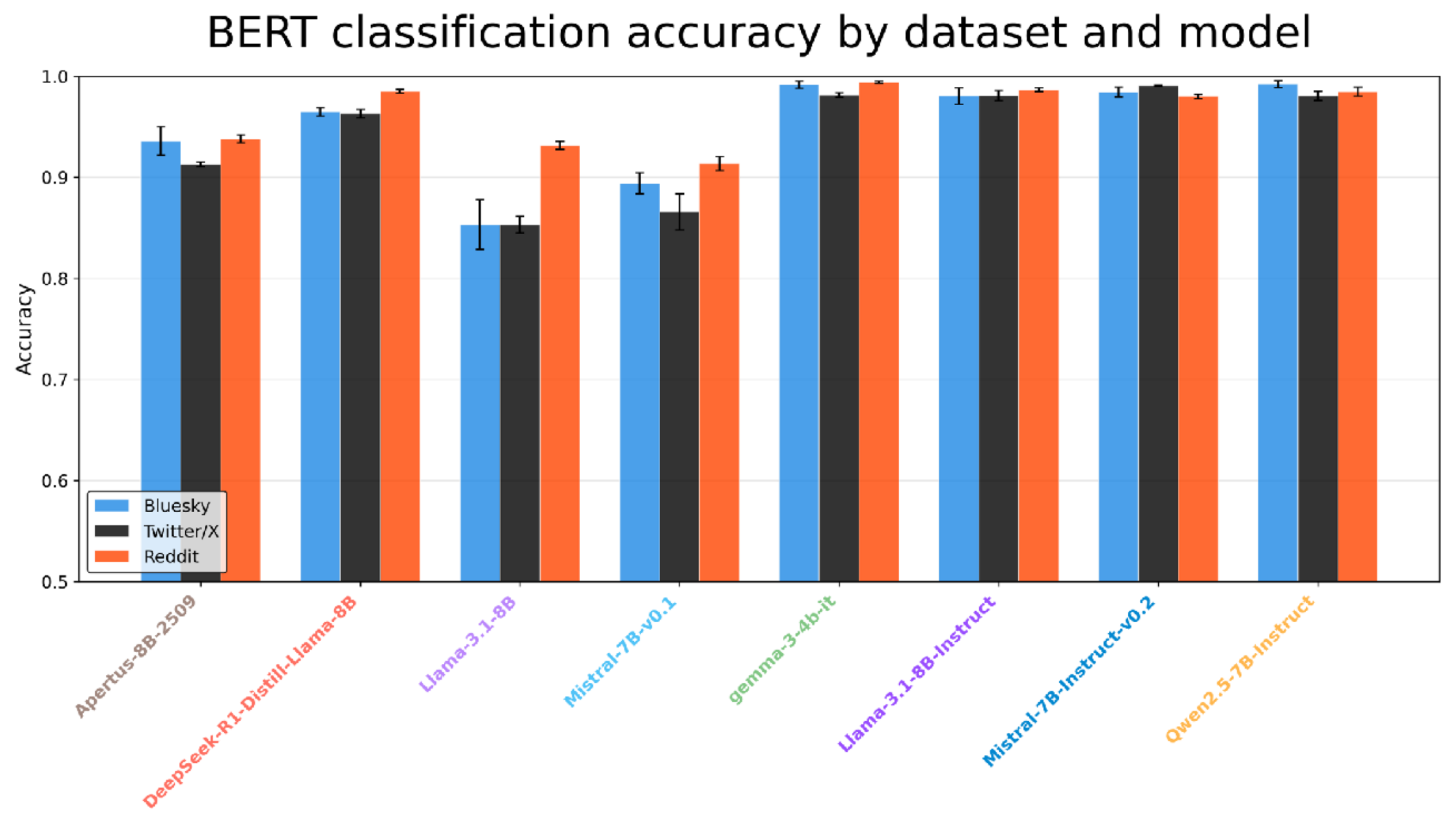
BERT-trained model | Cosine Similarity | Random Forest model | Empath Library

STATE-OF-THE-ART RESULTS:

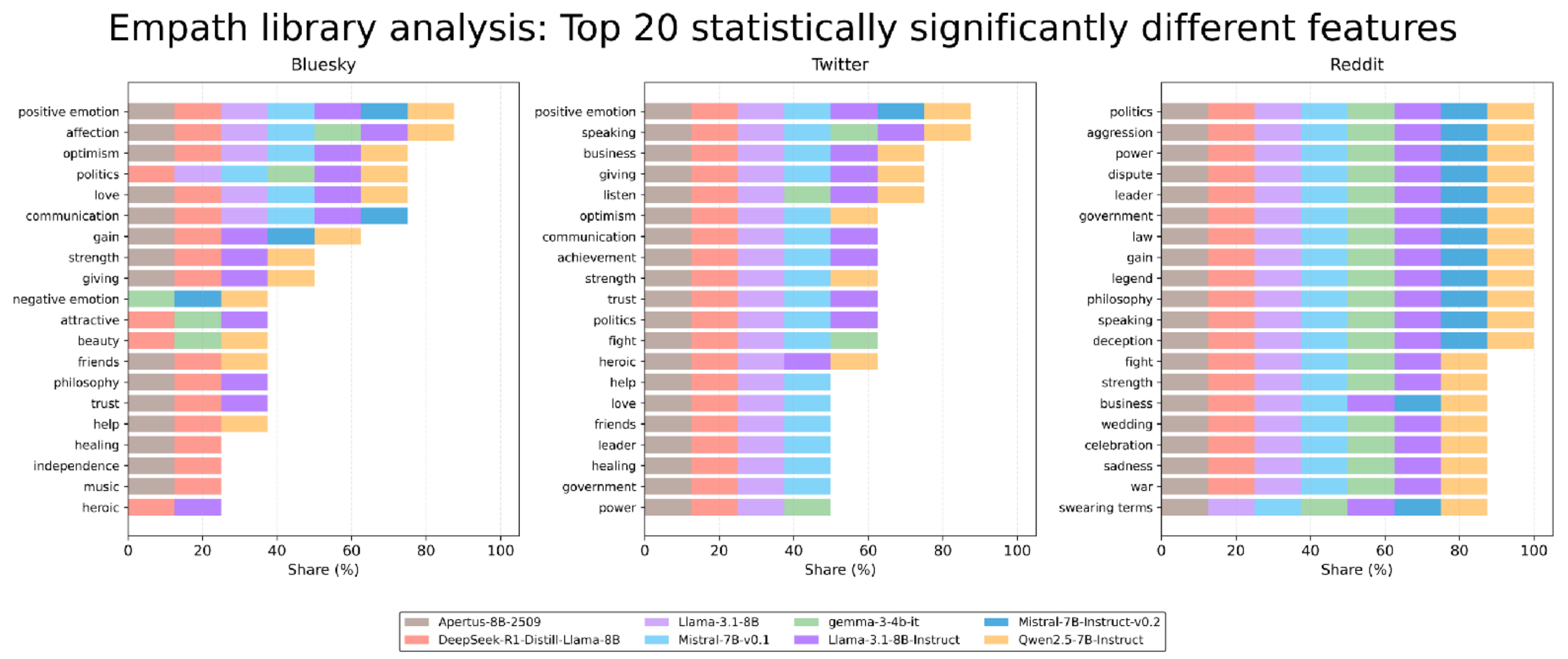
We rigorously tested state-of-the-art methods (consisting of **prompting with persona description**) on **3 data-sets** and **8 different models**. In total, we studied **800+ social media users**, generating replies to **20 real messages per user**.



Low cosine-similarity between AI-generated text and corresponding ground-truth text



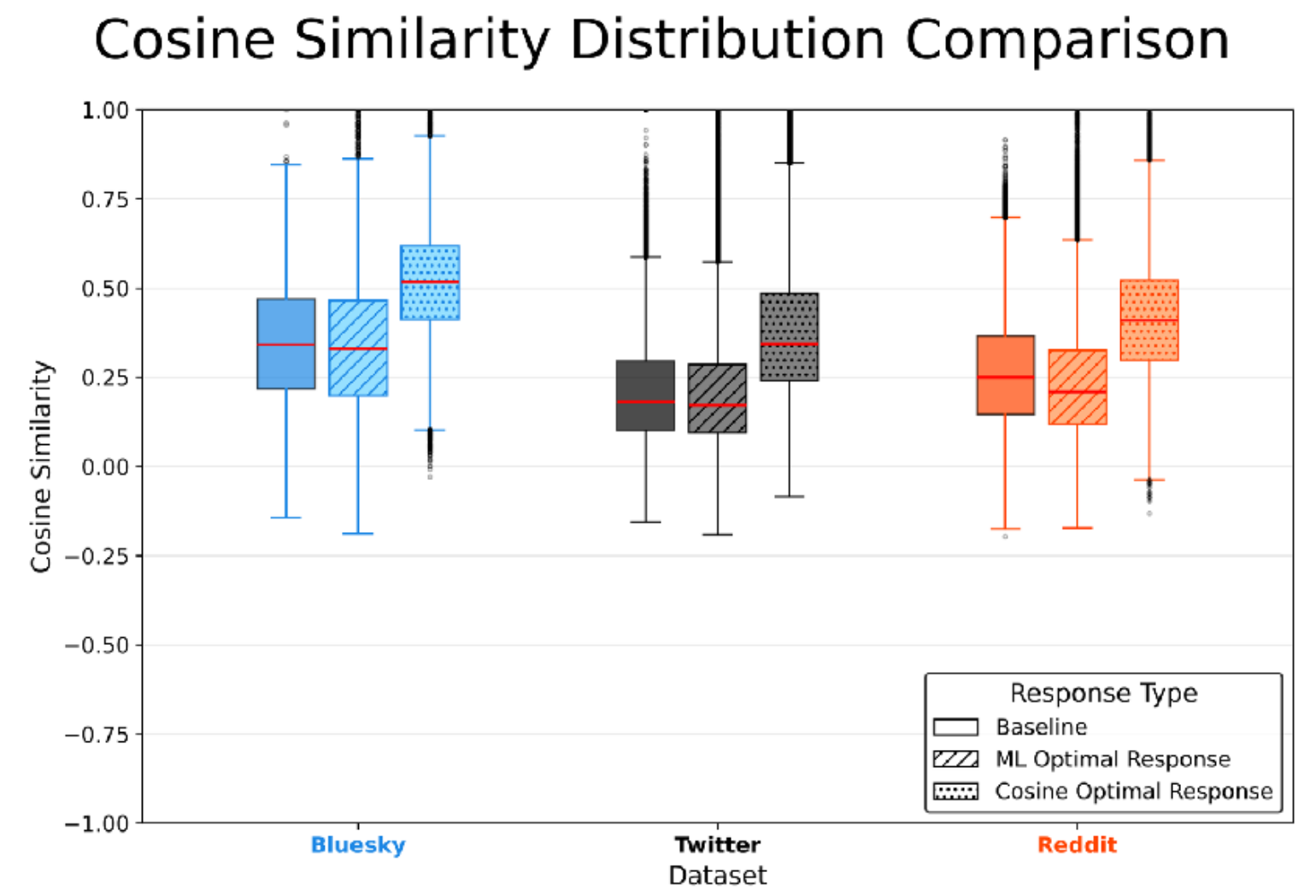
BERT-trained classifiers obtained high detection rates (accuracy larger than 85% on all models and datasets)



AI-generated text differs in many semantic features

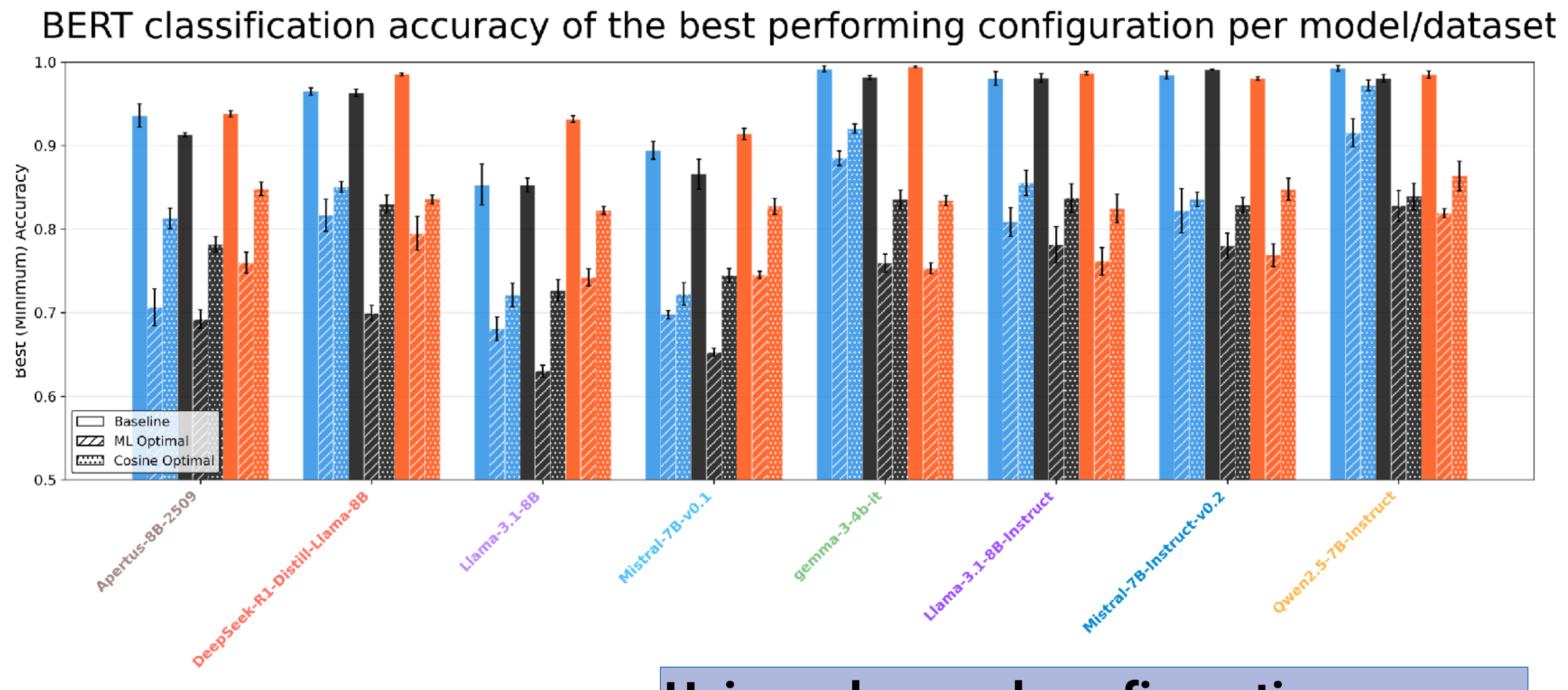
CALIBRATION RESULTS:

We then integrated (in consecutive steps) **stylistic examples**, **retrieved generated content**, and **fine-tuning**. For each AI reply, **we generate 20 alternatives**. Then we use (i) an **ML-judge** (based on several syntactic and emotional features, e.g., word count, average word length, sentiment, toxicity) and (ii) a **cosine-similarity judge** to select an optimal reply.

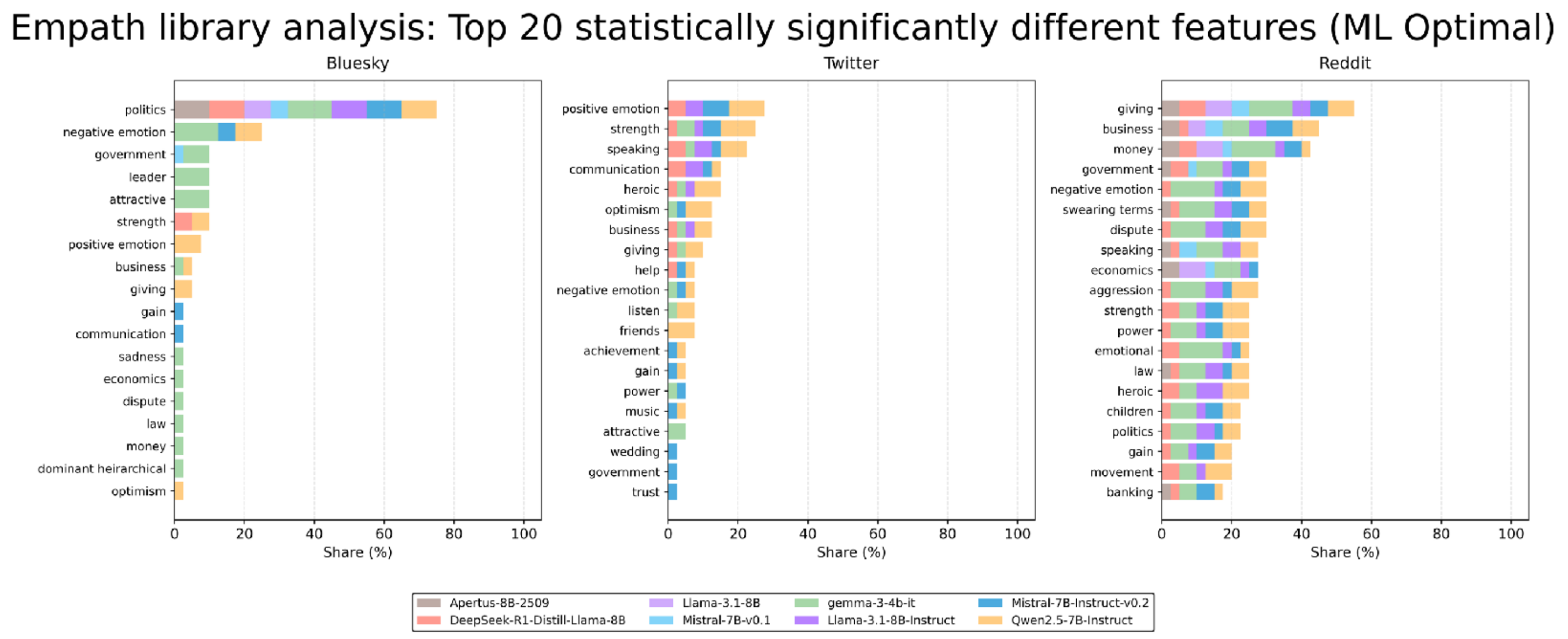


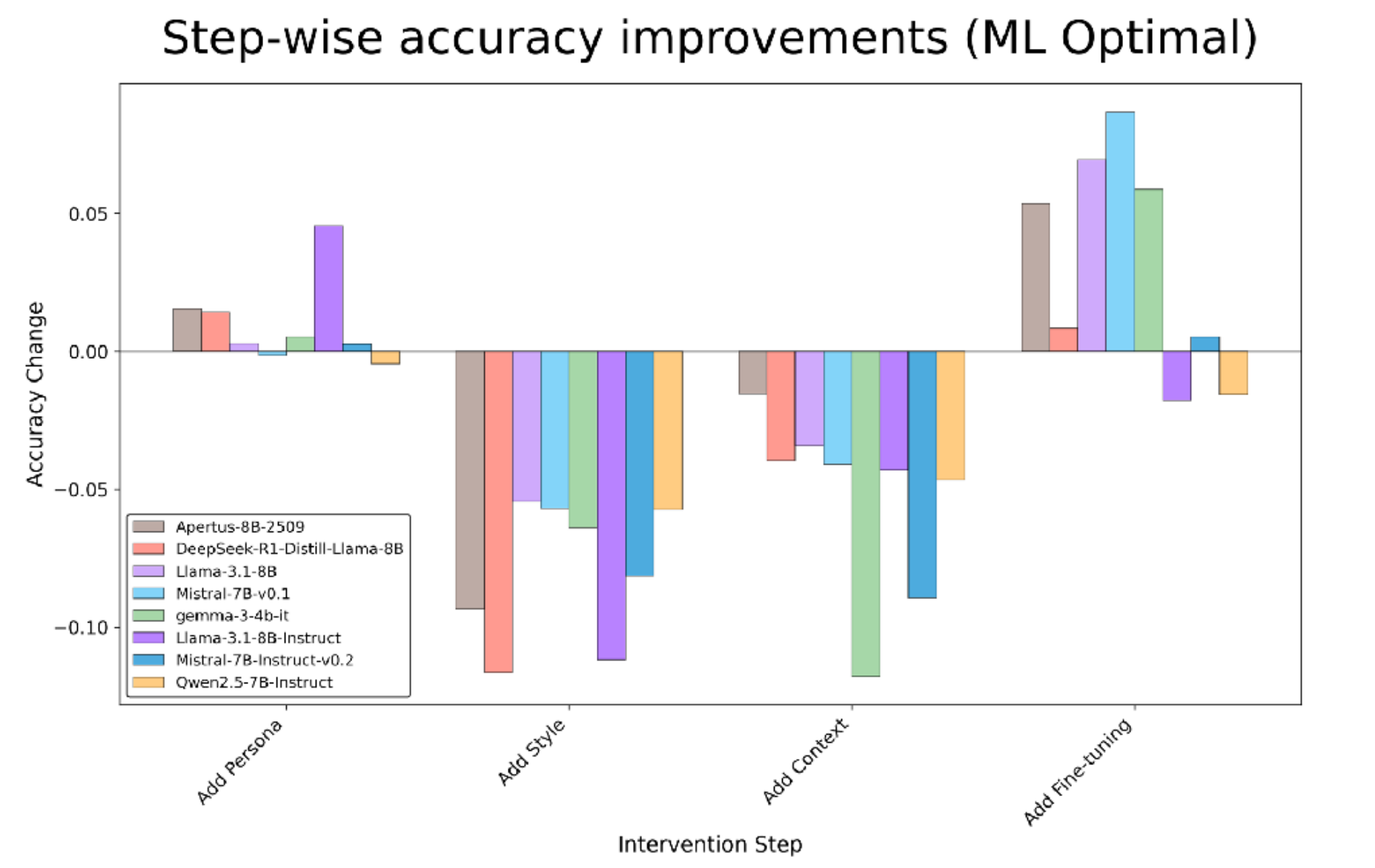
Optimising for cosine-similarity gives a marginal improvement

ML-optimal choice greatly reduced the number of features



Using advanced configurations (mostly SE + CR, no fine-tuning) pushes down the accuracy of the BERT-classifier, especially for Llama 3.1-8B, Mistral-7B-v0.1, Apertus 8B (and using the ML-optimal choice)





TAKE-AWAY:

Standard prompting methods fail spectacularly: all eight models with persona descriptions produces easily detectable text across all platforms and metrics.

TAKE-AWAY:

Optimised models generally reduce the gap between AI and human generated replies. Yet, **LLMs are still 'too nice' to be human**, with **more positive and less toxic content, and distinct syntactic features**.